

C7-C8

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Instruire asistată de calculator C7-C8

**Realitate augmentată (in engl. Augmented Reality)
si realitate virtuala (in engl. Virtual Reality) in educație**

- Generalități, exemplificări.
- Introducere in utilizarea minimala a unei platforme ce permite gratuitate temporara la dezvoltarea de aplicații AR pentru mobile.
- Prezentarea si experimentarea unui experiențe AR si VR.

Ce vom face?



01 Descoperim

Ce stii despre AR? Dar VR?

02 Invatam

Incearca si tu.

03 Fixam

Fixare prin discuții în grupuri de lucru (breakout rooms/face-to-face)

04 Recapitulam la final

Realitate augmentată

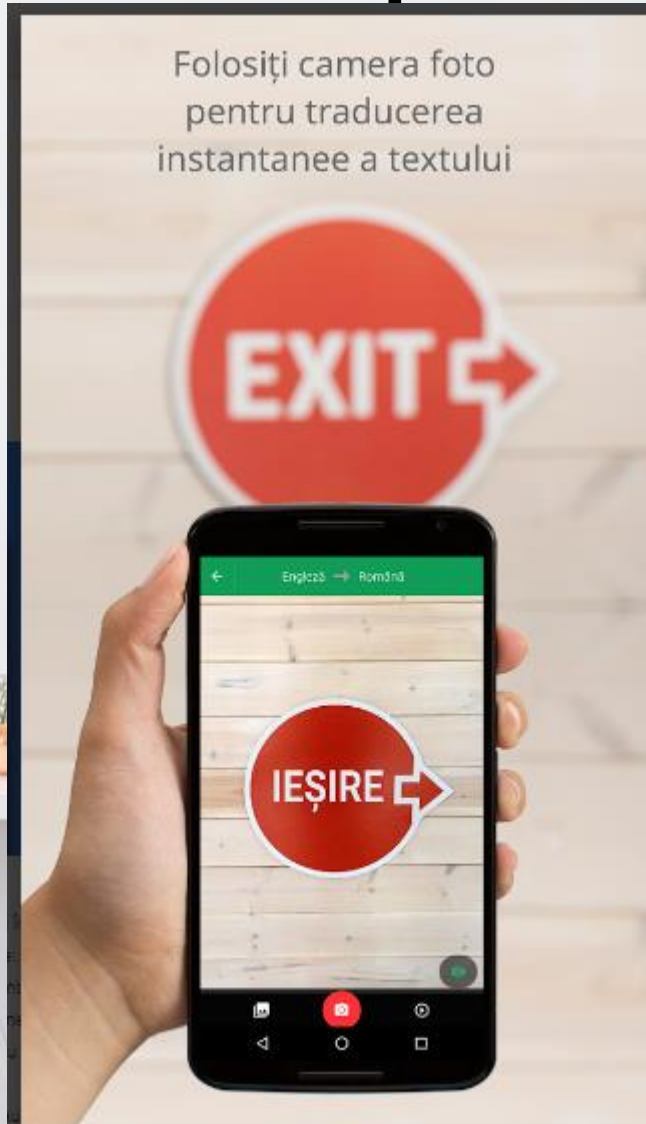
Realitatea augmentata (engl. Augumented Reality)

“AR modifies the user's actual physical surroundings by overlaying virtual elements (images, videos, virtual items;”

Carlos Flavián, Sergio Ibáñez-Sánchez, Carlos Orús, The impact of virtual, augmented and mixed reality technologies on the customer experience, Journal of Business Research, Volume 100, 2019, Pages 547-560, ISSN 0148-2963, <https://doi.org/10.1016/j.jbusres.2018.10.050>.
(<https://www.sciencedirect.com/science/article/pii/S0148296318305319>)



Exemple de aplicatii AR



- **Google Translate**
- **Pokemon GO – Nintendo**
- **Ikea mobile APP** (<https://www.designrush.com/best-designs/apps/ikea-place>)
- **Aplicatia Disney pentru cartea de colorat**
(<https://www.youtube.com/watch?v=SWzurBQ81CM>)
- **Microsoft Hololens for a music instruction in a Virtual/Augmented Environment**
- **MusicLens - A Microsoft HoloLens App for Musicians**
<https://www.youtube.com/watch?v=4I1RrFp0KWc>
- **Katana XR**
<https://world.magicleap.com/en-us>
- **How to Make an XR Guide in Under 5 Minutes with No Coding Knowledge /Katana Pro Create Tutorial**
<https://youtu.be/5ZcwhlCxwp4>

Exemple de platforme pentru dezvoltare AR

Aurasma = HPReveal, now ending

“Aurasma was started by UK software company Autonomy, which was acquired by Hewlett-Packard Enterprise (HPE) in October 2011 in a deal that valued Autonomy at \$11.7 billion (£7.4 billion). After the acquisition, the company operated as HP Autonomy.” (Sursa: <https://www.auganix.org/hud/aurasma/>)

Lens Studio

- <https://lensstudio.snapchat.com/news/>

ARCore from Google (free)

- <https://developers.google.com/ar>

ZappAR (a version where no coding is required)

- <https://zap.works/>

XR+ (a platform where no coding)

- <https://xr.plus/>

Unity

- unity.com/

KATANA Magic Leap

<https://developer.magicleap.com/downloads/magicversesdk>



Exemple de aplicatii AR

Aplicatie pentru Constructii de masini si inginerie

<https://www.youtube.com/watch?v=WmHJgQkoFB0>

Future Sight AR (Magic Leap, Katana XR app)

<https://youtu.be/SgobUvcFBf8?t=58>

Carte digitala

<https://www.youtube.com/watch?v=mwrHkbLCHP0>

Augmented & Virtual Reality(AR/VR) Based Interactive Engineering Training(Digital Book)

Manifest AR - Augmented Reality ROI for Manufacturing Machine Operator Training

<https://www.youtube.com/watch?v=0iUoAO7Z-io>

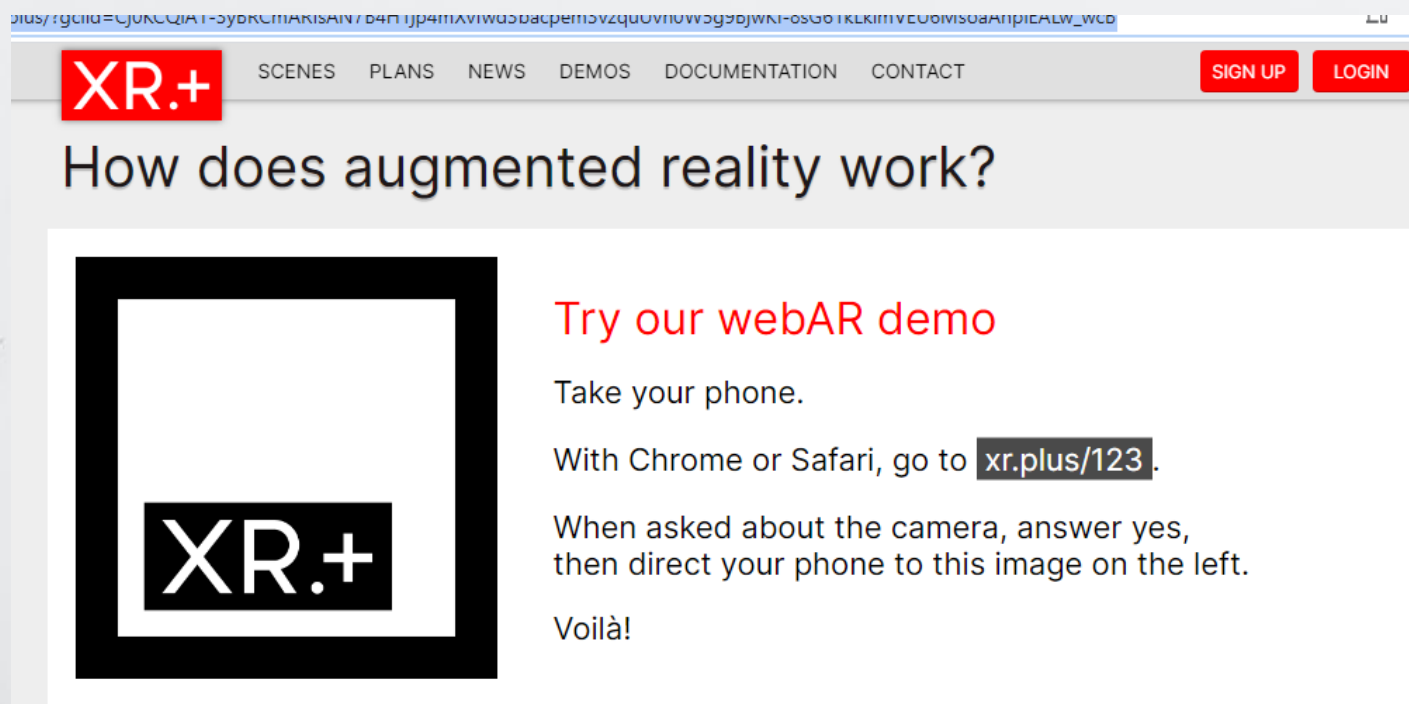
Augmented Reality Maintenance Training

https://www.youtube.com/watch?v=G1xNd_n1QyE



More: Augmented Reality

- Acum hai sa incercam un exemplu de Realitate augmentata 😊.



SAU

Du-te la adresa <https://xr.plus/>.

Cauta Quick demo.

Scaneaza codul QR cu QR Scanner de pe telefonul mobil.

Urmează indicatiile. Pune pe chat printscreen de pe telefonul mobil, pe măsură ce accesezi experienta AR.

Creează o aplicație simplă folosind ZapAR web based.

- Materialul pentru crearea unei aplicații AR folosind ZapWorks este accesibil

AICI.

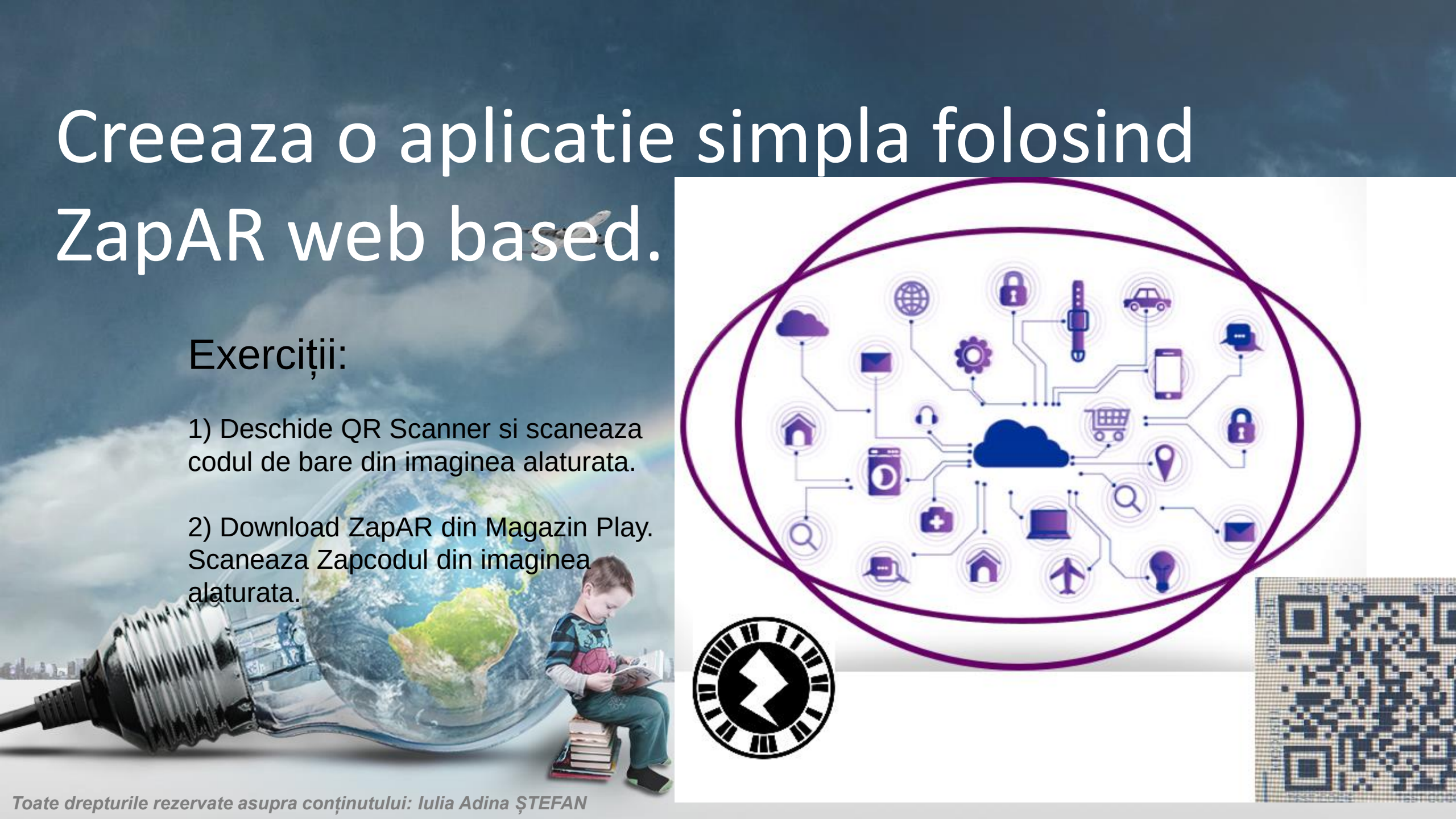
- Interfata a putut suferi modificări, se vor folosi, în acest caz, printscreen-urile livrate în chat, în timpul exercițiului.
- Atenție la definirea unei scene, realizarea tranziției între scene, adăugarea de efecte pentru butoane



Creeaza o aplicatie simpla folosind ZapAR web based.

Exerciții:

- 1) Deschide QR Scanner si scaneaza codul de bare din imaginea alaturata.
- 2) Download ZapAR din Magazin Play. Scaneaza Zapcodul din imaginea alaturata.



Toate drepturile rezervate asupra conținutului: Iulia Adina ȘTEFAN

1) Deschide QR Scanner si scaneaza codul de bare din imaginea alaturata.

2) Download ZapAR din Magazin Play.
Scaneaza Zapcodul din imaginea alaturata.



Realitate virtuala



Realitate virtuala (engl. Virtual Reality):

“the VR environment is programmable and everything in VR is manufactured.”*

*Z. Zhang, B. Cao, J. Guo, D. Weng, Y. Liu and Y. Wang, "Inverse Virtual Reality: Intelligence-Driven Mutually Mirrored World," 2018 IEEE Conference on Virtual Reality and 3D User Interfaces (VR), 2018, pp. 735-736, doi: 10.1109/VR.2018.8446260

“VR technology is a kind of artificial environment supported by computer technology, which has functions to provide the users with perceptions of vision, hearing, touch, etc.”

“There are three characteristics of VR technology, immersion, interaction and imagination. Among them, immersion refers to the user can feel his real existence in the virtual environment. Interaction means that the objects in virtual environment can be operated and feedback to natural environment. Imagination indicates that the living environment or the hero engaged actions provide the users with infinite imagination space. In such a virtual environment, people can take next activities according to their thinking.”**

**S. Li, "Research on Application of Virtual Reality Technology in Teaching and Training," 2009 Second International Conference on Computer and Electrical Engineering, 2009, pp. 77-80, doi: 10.1109/ICCEE.2009.22



Obiectivele tehnologiei VR in educație și formare

“The roles of VR technology plays in the teaching and training can be concluded as following:

1.Enhance learning effect. The realistic scene that constructed by VR technology can provide interaction with real-life, the immersion provided by which can enhance good learning effect.

2.Reduce capital investment. Application of VR technology in teaching and training can solve the problems of field lack, climate impact and material loss that of real equipment training. VR technology can completely break the restrictions of space and time. Therefore, the application of VR technology can significantly improve the efficiency of teaching and training, so that good benefit can be achieved.

3.Safe and reliable. Exertion of VR technology in military teaching and training can ensure safety and effectively avoid casualties. According to related statistics, in the real man and real equipments training of close-to-warfare in high-tech condition, the casualties have increase trend. The teaching and training based on VR technology can be actualized in the assumed real environment, so the risk of military training can be reduced and the casualties in training can be effectively avoid.

4.Be re-usable and convenient to update. A set of VR system can be used by many participants and trainees. In addition, with the change of teaching content, update of armament equipment type or change of operational strategy, the VR system can also be updated.”*



Exemple de aplicatii AR si /sau VR

ClassVR: VR & AR in education

VR In Education & Industry

"ClassVR is a versatile platform using the power of Virtual and Augmented Reality for education and training from the classroom to the boardroom."

(sursa <https://www.classvr.com/>)

Resurse de invatare in Realitate virtuala

- <https://www.classvr.com/the-gunpowder-plot-virtual-reality-teaching-resources-blog/>

360° Underwater National Park | National Geographic

- <https://www.youtube.com/watch?v=v64KOxKVLVg&list=PLRzcDoB0iZQWYfpYUSW4xl2nHxtbCgYYS>

360° Great Hammerhead Shark Encounter | National Geographic

https://www.youtube.com/watch?v=rG4jSz_2HDY&list=PLRzcDoB0iZQWYfpYUSW4xl2nHxtbCgYYS&index=2

Human Digestive System in VR!!! | Education in 360

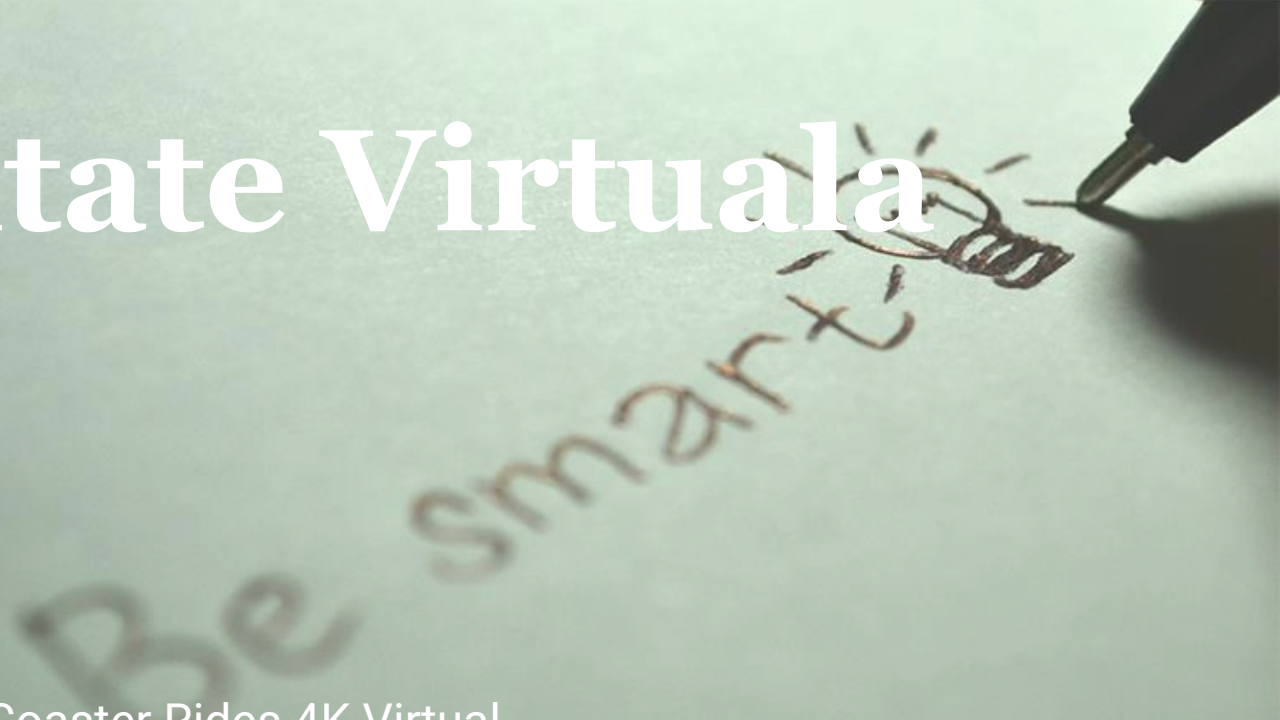
<https://www.youtube.com/watch?v=r8-6Z1Efgzg>

360° FEAR OF HEIGHTS! FALL AND TITAN EATS YOU! VR Experience

https://www.youtube.com/watch?v=2OS_eqTieh8



Jocuri de Realitate Virtuala



VR Games

- Astro Bot Rescue Mission
- Second Life VR platform
- 3D VR Video
- VR 360 Video of Top 5 Roller Coaster Rides 4K Virtual Reality
- Best VR 360 VIDEO (Virtual Reality)

Hardware to use: Rokoko

Ce este Samsung GearVR?

EXERCİȚIU

ATENȚIE–INTRĂ

ȘI LA EXAMEN

Breakout Rooms



Creați în echipă un exemplu în platforma ZapWorks, care să definească două scene, în scena 1 fiind două butoane (unul pentru acces listă de proiecte, unul pentru acces la scena 2), și o a doua scenă, cu 3 imagini inserate, fiecare imagine având asociat un link pentru acces un site sau videoclip Youtube.”





Recap

Ce instrumente am folosit azi? Interactiune pe jamboard.

01

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Mulțumesc pentru atenție!

